

BERLIN TEGEL, Germany

WMO: 103820

Lat: 52.5644N

Lon: 13.3089E

Elev: 121

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	19.0	5.2	7.2	16.1	10.6	9.5	21.9	26.3	45.2	24.1	43.7	8.5	70	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	88.2	66.6	84.5	65.5	81.0	64.6	69.3	81.6	68.0	79.9	66.6	77.6	8.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	96.2	72.7	64.3	90.6	72.3	62.5	85.1	71.1	33.4	81.6	32.3	80.4	31.1	76.8	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(4) 21.4	18.9	16.7	DB	8.9	94.6	6.8	3.5	4.0	97.2	0.0	99.2	-3.8	101.2	-8.8
(5)			WB	8.1	73.1	6.7	1.5	3.2	74.2	-0.7	75.1	-4.5	75.9	-9.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	33.4	34.9	41.3	51.5	58.2	64.9	68.6	67.0	60.3	51.2	43.1	37.4
	DBStd	14.18	8.74	7.58	7.84	7.53	7.62	6.80	6.34	5.84	5.87	6.42	6.65	7.72
	HDD50	1982	514	424	282	71	13	0	0	0	1	64	222	391
	HDD65	5473	979	844	735	407	238	84	34	44	167	428	658	855
	CDD50	2370	0	0	12	116	266	446	577	527	309	102	14	1
	CDD65	385	0	0	0	2	27	79	146	105	25	1	0	0
	CDH74	3318	0	0	0	27	253	727	1311	817	178	5	0	0
	CDH80	1032	0	0	0	3	62	230	452	256	29	0	0	0
(14) Wind	WSAvg	8.0	9.5	8.7	8.7	8.0	7.7	7.4	7.2	6.9	7.1	7.4	8.3	9.3
(15) Precipitation	PrecAvg	24.1	1.9	1.3	1.3	1.4	2.4	2.5	2.8	2.6	1.7	1.7	1.9	1.9
	PrecMax	35.7	4.3	2.9	3.8	3.2	5.8	7.1	8.0	6.3	4.2	5.4	3.9	5.1
	PrecMin	16.9	0.6	0.1	0.4	0.1	0.3	0.1	0.3	0.3	0.2	0.2	0.3	0.4
	PrecStd	5.0	0.9	0.8	0.7	0.9	1.3	1.6	1.8	1.3	0.9	1.2	0.9	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	55.0	66.3	77.2	86.3	91.3	94.8	91.8	84.2	73.2	61.1	55.6
		MCWB	48.5	45.3	51.5	57.8	64.5	68.4	68.0	68.3	65.0	60.6	56.6	50.4
	2%	DB	50.4	50.3	60.8	73.0	80.6	86.0	89.2	86.4	78.9	68.0	57.3	52.1
		MCWB	47.2	45.3	49.8	56.2	62.4	65.5	66.6	67.4	63.4	58.5	53.1	48.5
	5%	DB	48.0	48.2	56.9	69.6	75.5	82.2	85.6	82.2	75.2	64.4	55.0	50.0
		MCWB	45.3	44.0	48.0	54.3	59.6	64.5	66.1	65.5	61.9	56.9	50.8	46.8
	10%	DB	44.9	45.0	53.2	65.9	72.0	77.4	80.9	78.5	71.3	61.0	52.0	47.9
		MCWB	42.6	41.1	46.2	53.0	58.0	62.7	64.7	63.8	60.2	55.5	48.6	45.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	48.9	54.2	59.8	67.2	71.1	70.9	71.1	67.7	62.2	56.7	52.0
		MCDB	52.2	50.2	62.4	73.5	79.6	84.6	82.8	82.5	79.7	69.6	60.3	55.3
	2%	WB	47.6	45.9	51.2	57.5	64.6	68.2	69.3	69.1	64.9	59.8	53.5	49.2
		MCDB	50.4	49.9	58.1	69.0	77.0	80.1	81.8	81.2	74.8	66.2	56.8	51.2
	5%	WB	44.9	43.6	49.0	55.7	62.3	66.4	68.1	67.4	63.0	57.6	51.1	47.2
		MCDB	46.9	47.6	55.0	66.4	72.5	77.8	80.0	78.3	71.8	63.2	54.2	49.6
	10%	WB	42.6	41.2	46.8	53.8	59.9	64.4	66.7	65.7	61.2	55.8	49.4	44.8
		MCDB	44.7	44.6	52.1	64.3	69.3	74.9	78.2	74.6	69.5	60.5	51.9	46.9
(35) Mean Daily Temperature Range	MDBR		7.7	9.8	13.6	17.8	17.7	17.5	17.0	17.0	15.9	12.7	8.8	7.6
	5% DB	MCDBR	9.6	14.1	20.8	23.0	22.5	23.2	23.3	22.6	21.6	17.1	10.9	9.4
		MCWBR	8.1	10.0	11.8	10.9	9.6	8.9	8.1	8.7	9.6	9.8	7.9	7.6
	5% WB	MCDBR	8.7	13.0	18.1	21.1	20.6	20.4	19.6	18.7	18.3	15.6	10.3	9.3
		MCWBR	7.8	9.9	11.4	10.8	9.8	9.2	8.2	8.9	9.5	9.8	7.9	8.1
(40) Clear-Sky Solar Irradiance	taub		0.327	0.353	0.393	0.419	0.416	0.413	0.434	0.427	0.389	0.376	0.351	0.321
	taud		2.310	2.284	2.194	2.199	2.251	2.283	2.237	2.269	2.351	2.348	2.351	2.321
	Ebn at Noon		206	236	248	258	265	267	258	252	249	225	195	186
	Edh at Noon		20	27	36	41	41	40	41	38	31	26	19	17
(44) All-Sky Solar Radiation	RadAvg		198	419	747	1250	1548	1646	1541	1324	966	534	244	147
	RadStd		22	72	83	163	184	130	205	106	109	65	31	15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page